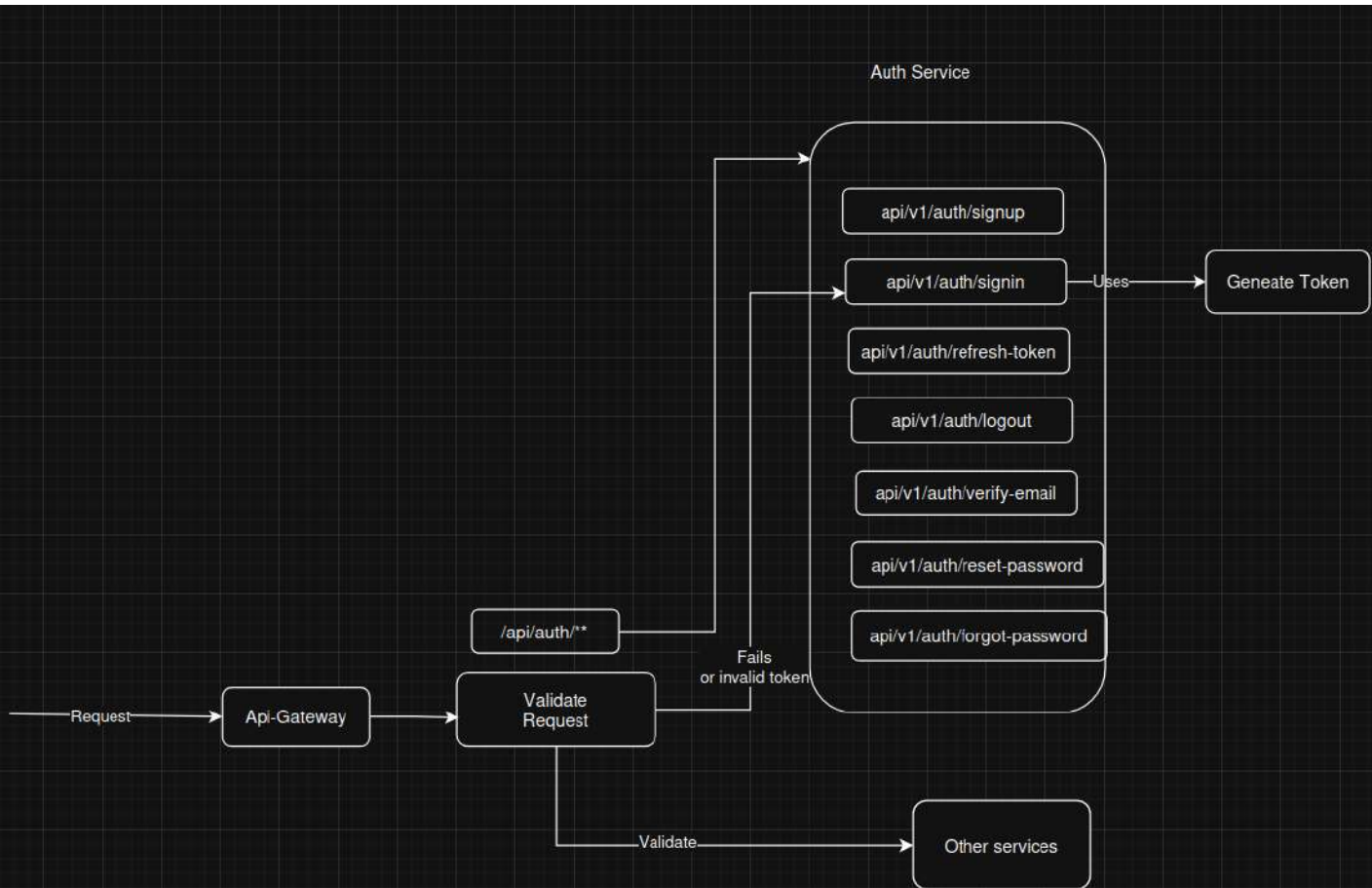
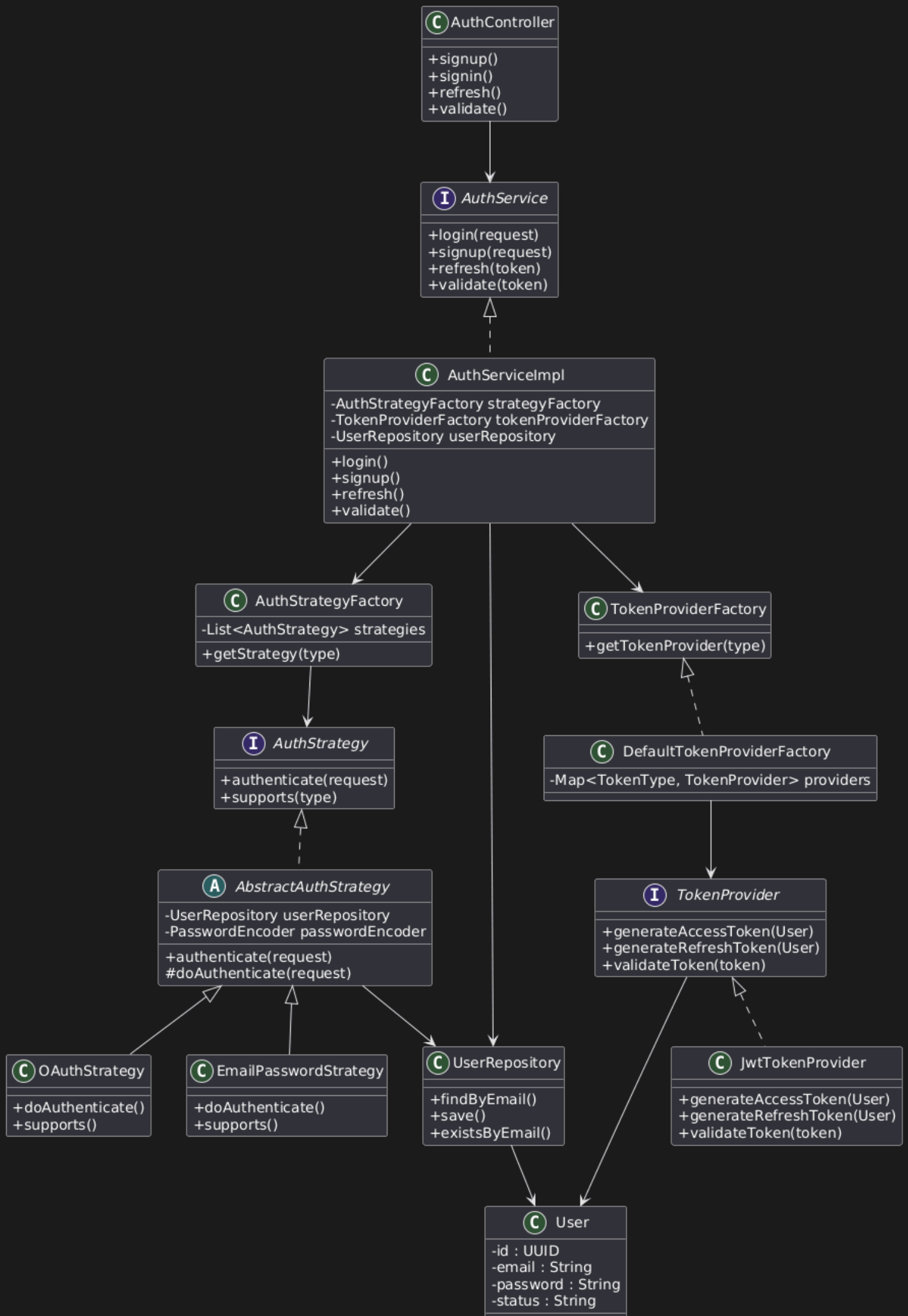


Auth Service Usecase Diagram

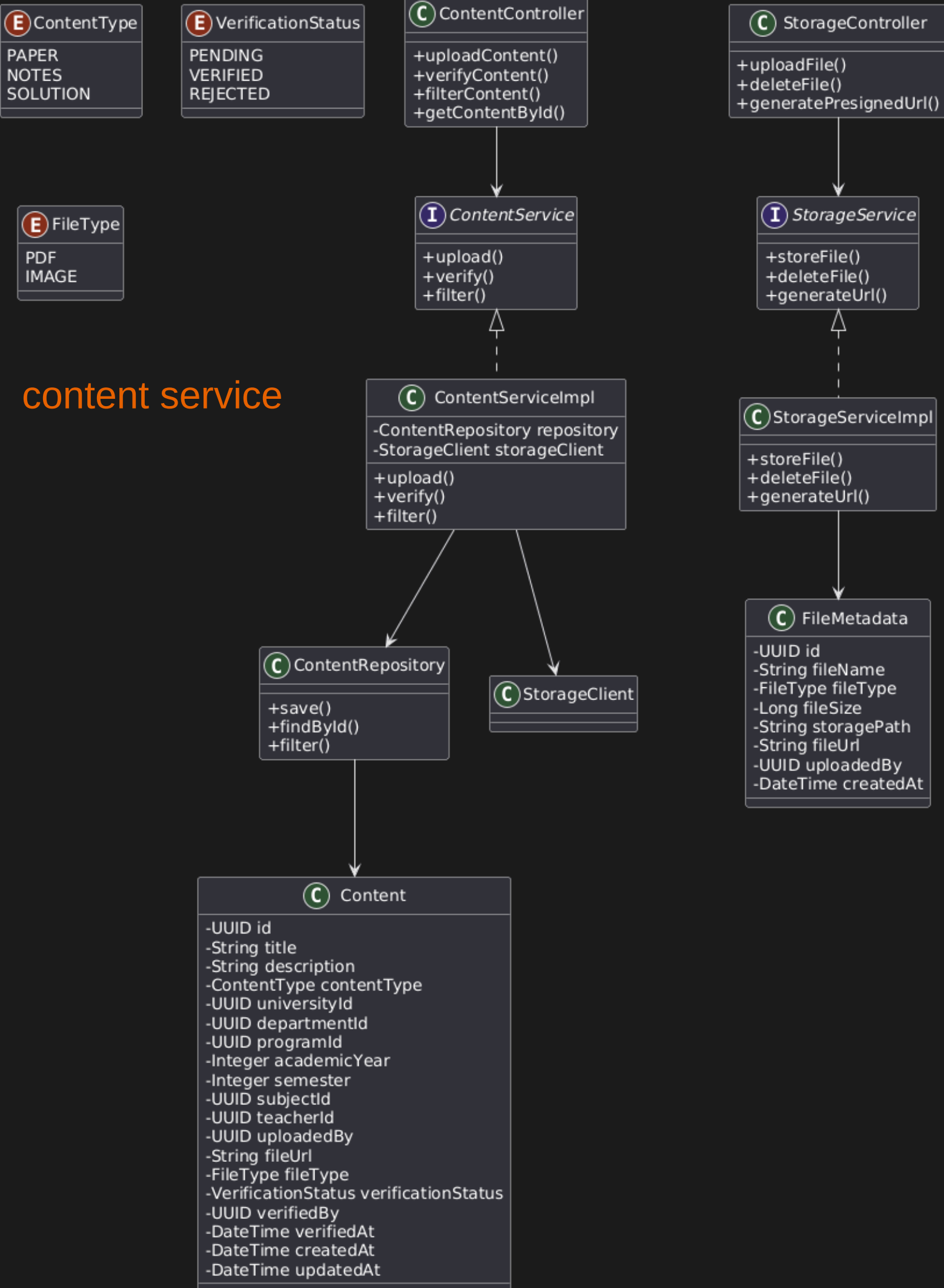


UML for Auth Service

Authentication Microservice - Complete LLD UML



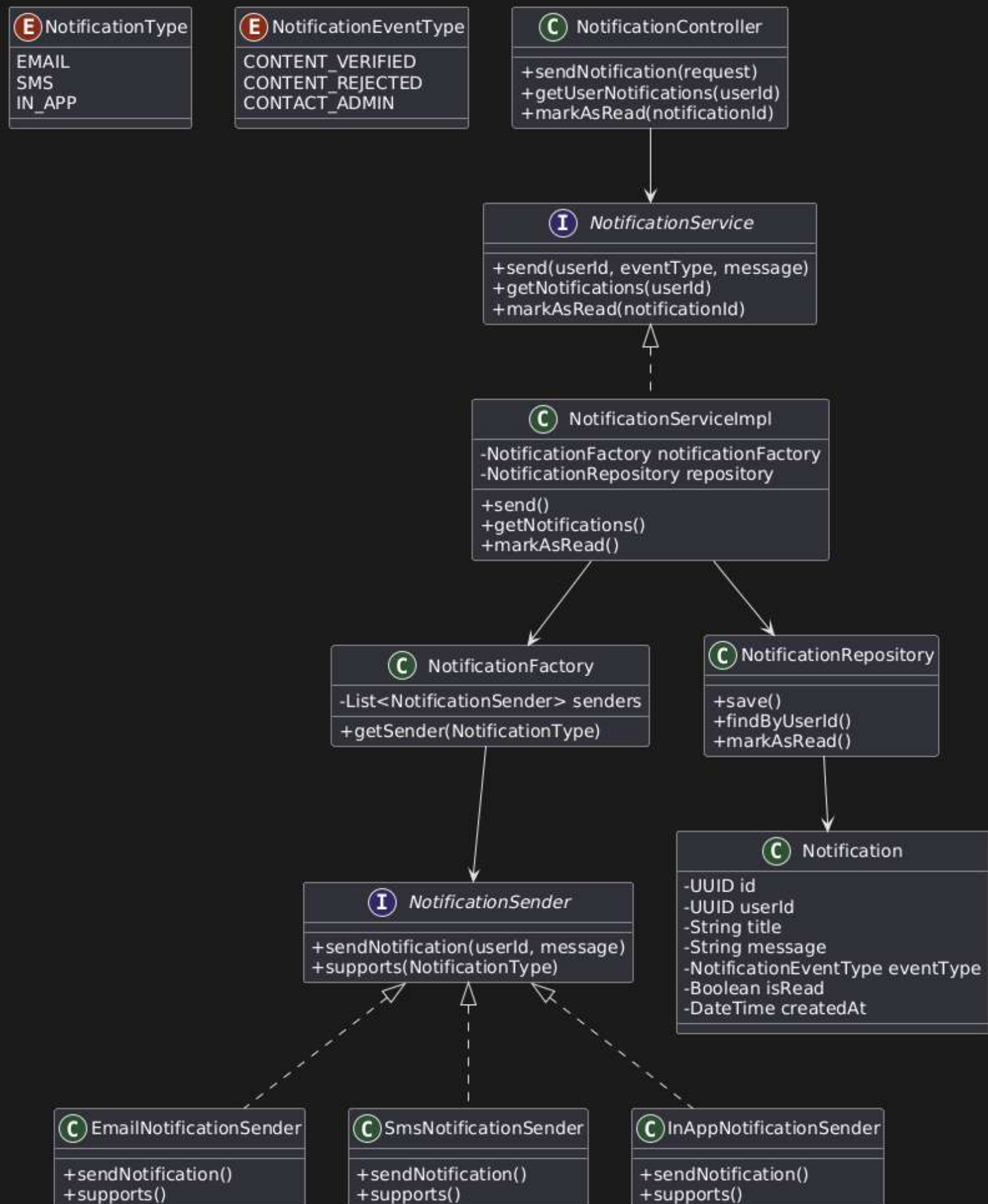
Content Service + Storage Service LLD



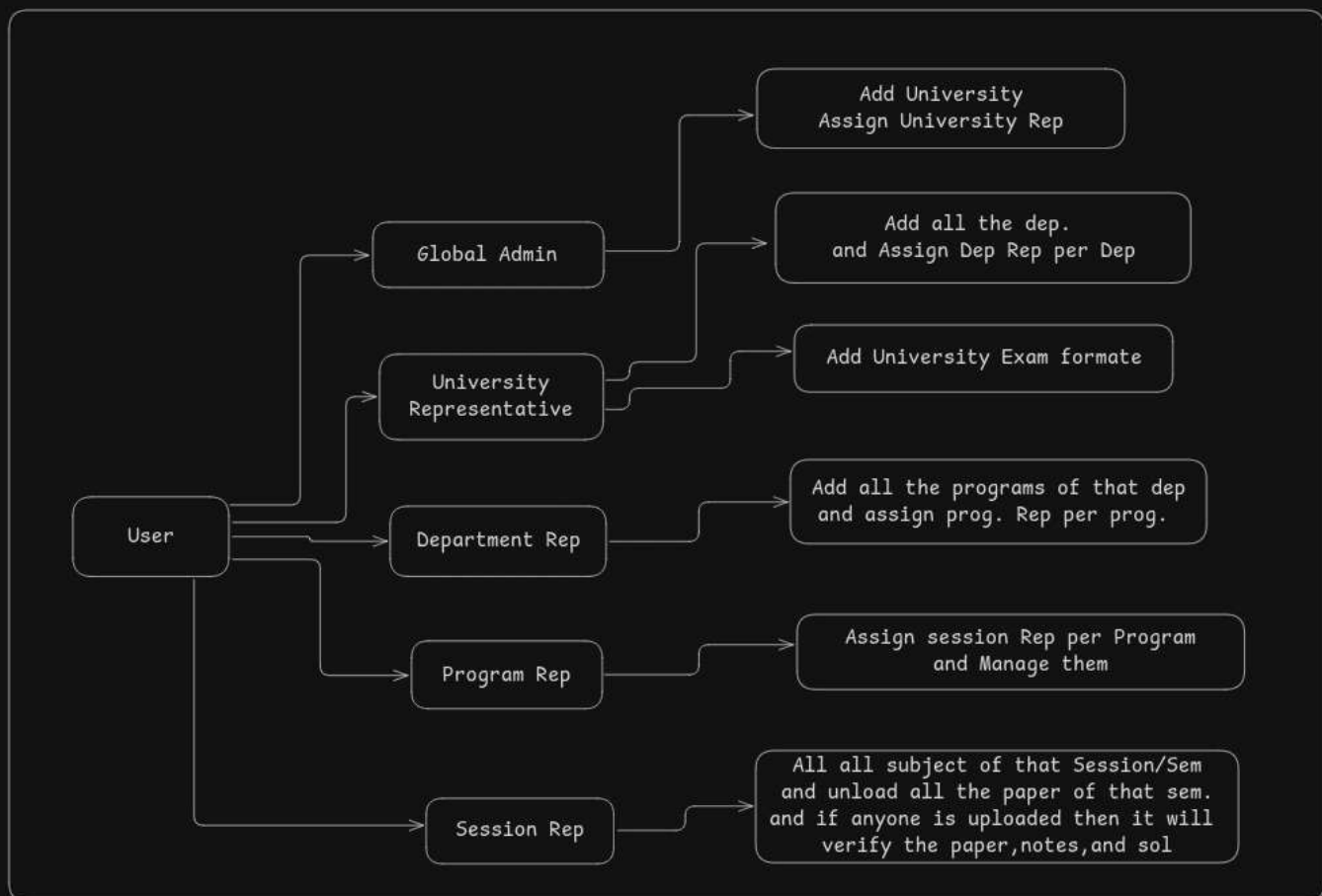
content service

Notification Service

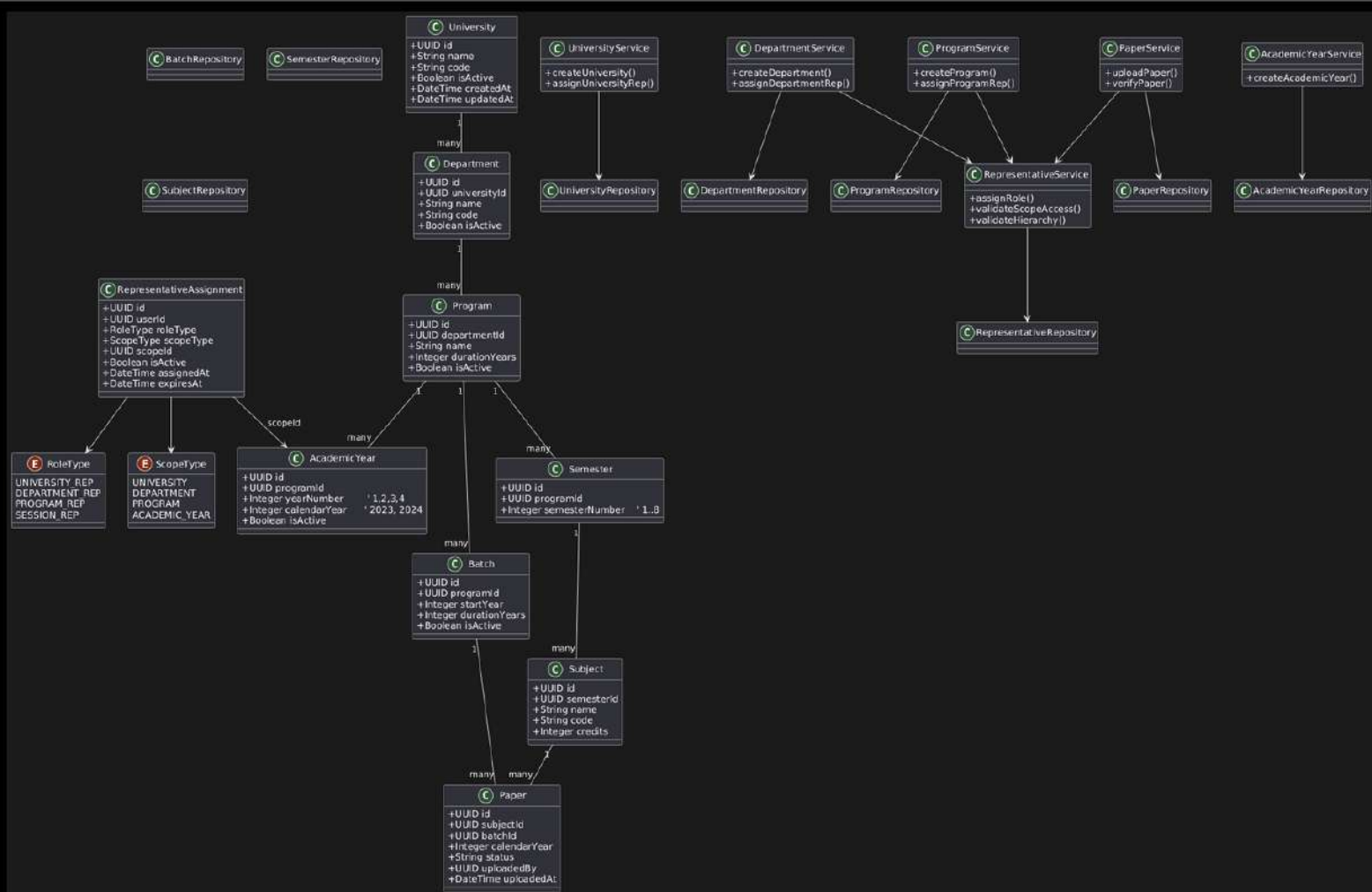
Notification Service - Complete LLD



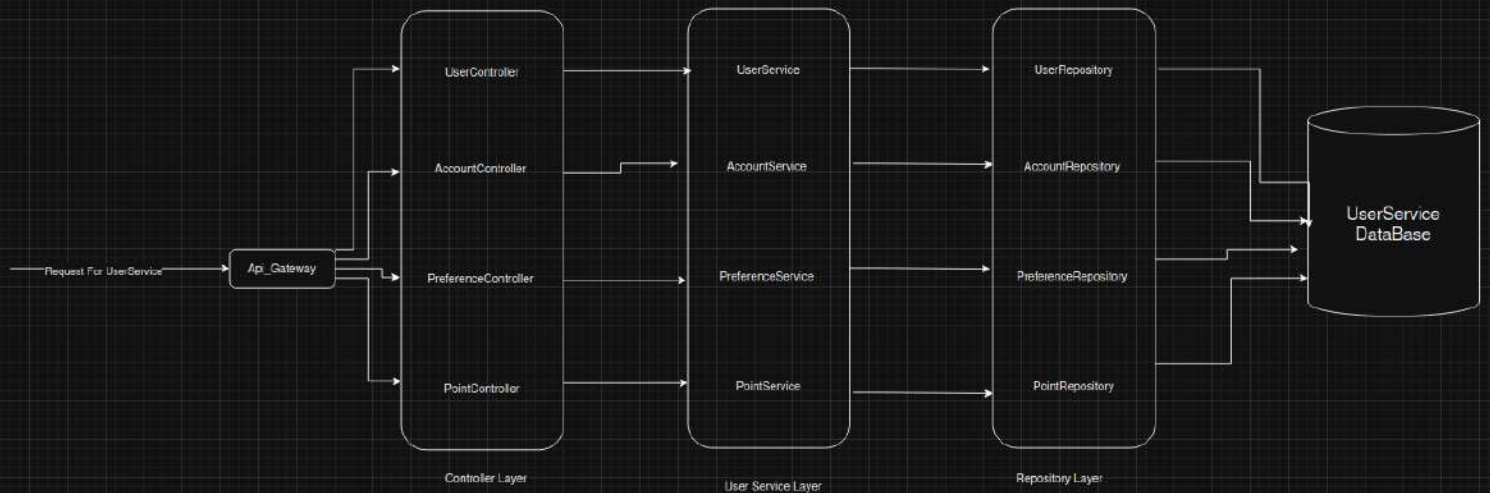
University Service Flow diagram



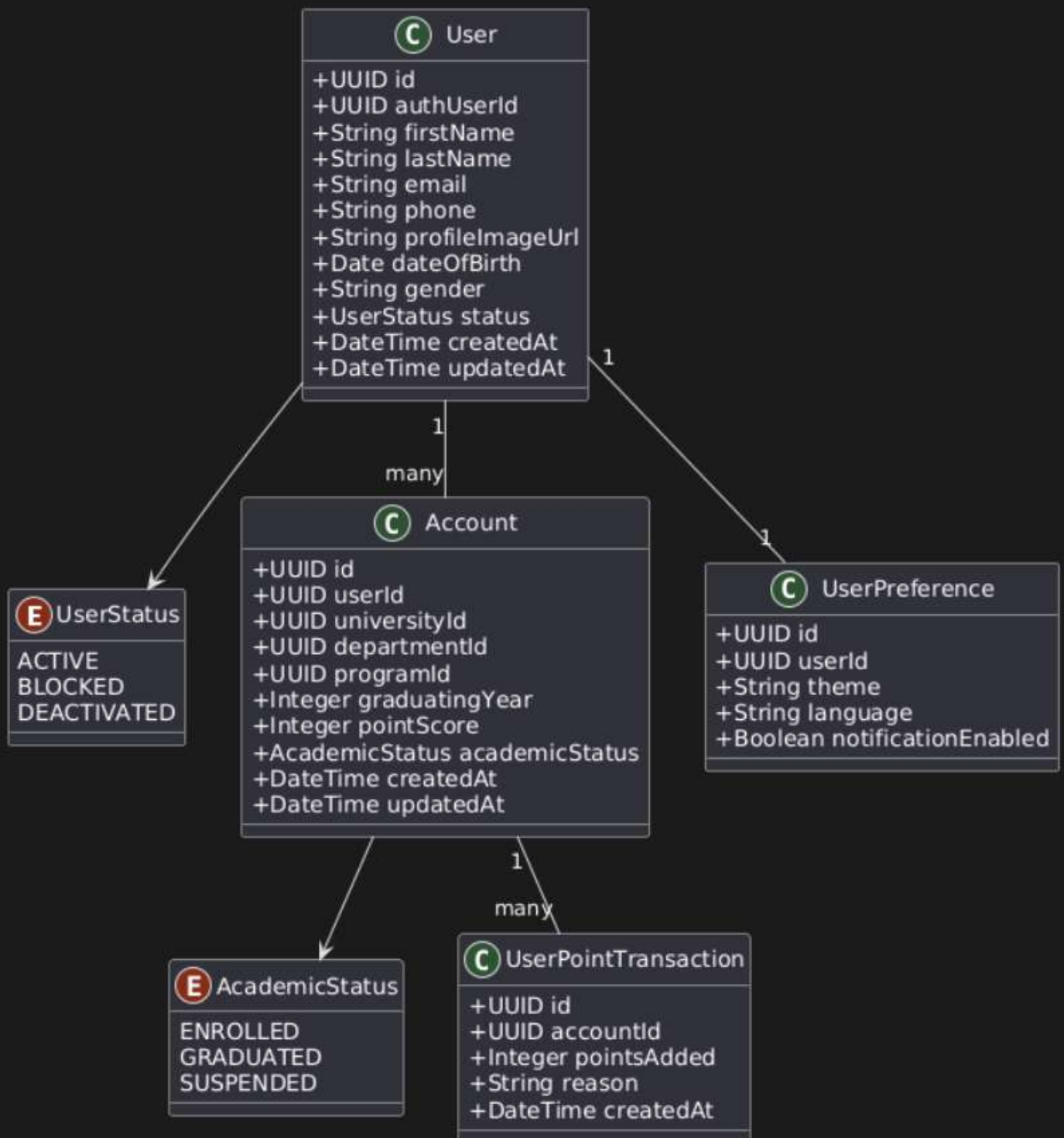
University Service UML



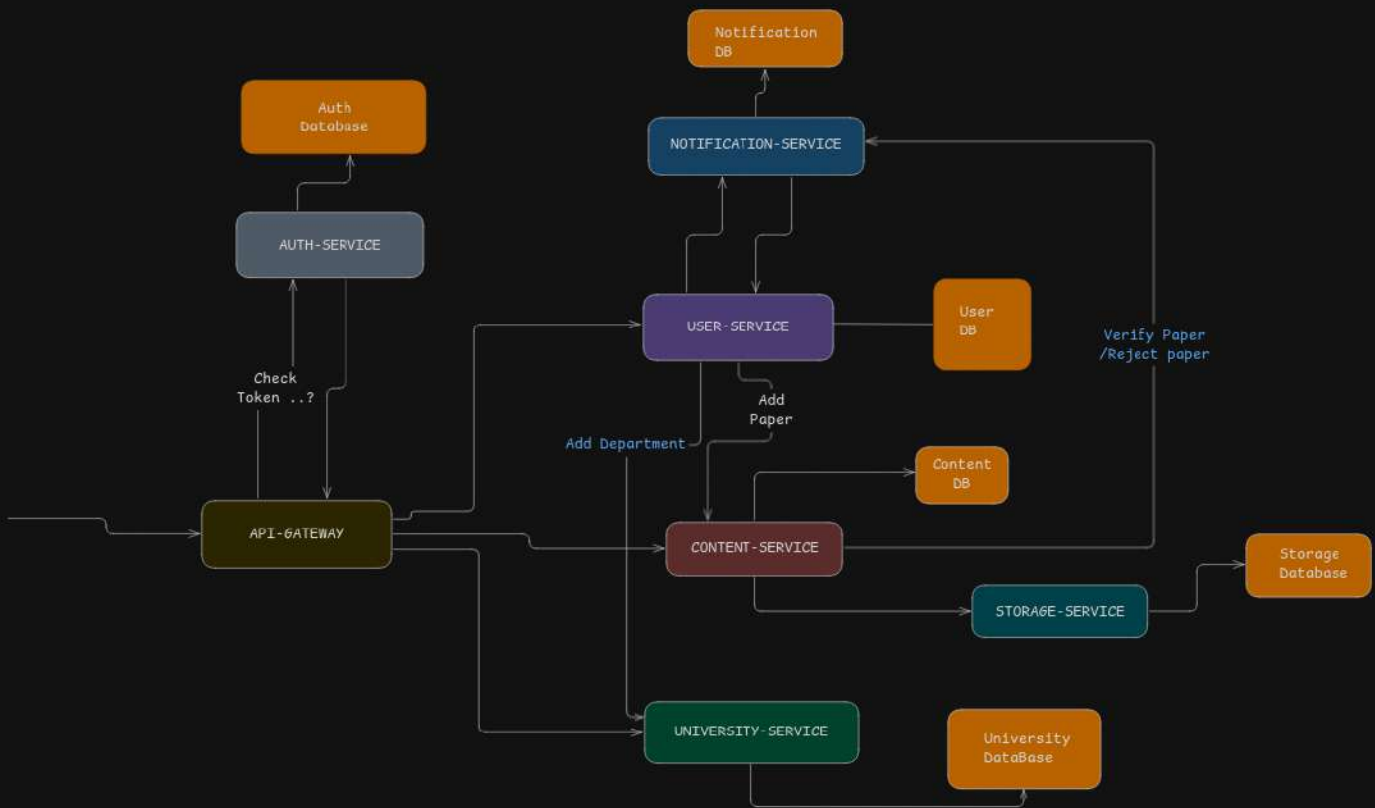
User service UseCase



User-service



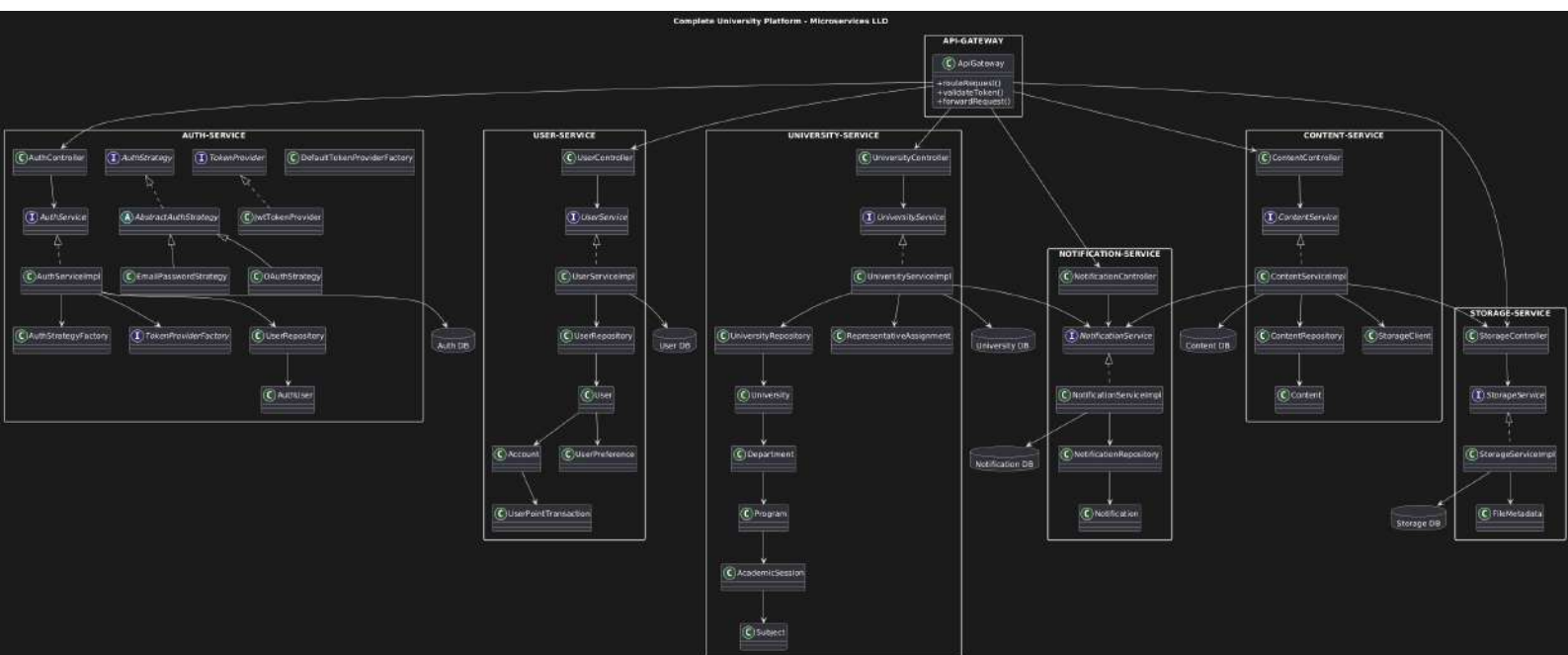
Whole Application



Use Case Diagram for Application



UML Diagram for whole Application



User Stories

Authentication Service

User Registration & University Selection

User Story: As a New User, I want to register an account using my email, username, and password, and select my University so that I can access relevant academic materials.

Acceptance Criteria:

The system must validate that the email is unique and follow a valid format.

The user must be able to select a University from the existing list managed by the University Service.

Passwords must be hashed using bcrypt before saving to the database.

A new user record is created with an isActive: false status until email verification is complete.

Upon successful registration, the user is automatically redirected to the OTP verification screen.

2. Email Verification (OTP)

User Story: As a Registered User, I want to verify my email address via an OTP so that my account is secured and activated.

Acceptance Criteria:

The system must generate a 6-digit OTP and send it to the user's registered email using the Resend service.

The OTP must have an expiration time (e.g., 10 minutes).

If the user enters the correct OTP, the isVerified status is updated to true.

The user should have an option to "Resend OTP" if the original email is not received.

3. Secure Login

User Story: As a Verified User, I want to log in with my credentials so that I can access my personalized dashboard.

Acceptance Criteria:

The system must verify credentials against the stored hashed password.

Upon successful login, the system must return a JWT (JSON Web Token) containing the user's userId and globalRole.

The system should return an error message for invalid credentials without revealing whether the email or password was wrong for security purposes.

4. Forgot & Reset Password

User Story: As a User, I want to reset my password if I forget it so that I can regain access to my account.

Acceptance Criteria:

The user enters their email to receive a password reset link or OTP.

The reset process must verify the user's identity before allowing a password change.

The new password must meet complexity requirements (e.g., minimum 8 characters).

Once changed, all previous active sessions/tokens for that user should be invalidated.

5. Secure Logout

User Story: As an Authenticated User, I want to log out of the platform so that my session is closed securely.

Acceptance Criteria:

The client-side token (JWT) should be deleted or cleared from local storage/cookies.

If using a token blacklist, the current JWT should be added to the blacklist until it expires.

The user is redirected to the login or landing page.

6. Role-Based Access (University Context)

User Story: As a Student, I want to see and upload papers specific to my selected University so that I can contribute to and learn from my community.

Acceptance Criteria:

The system must restrict "See Paper" and "Upload Paper" actions to the University ID selected during registration.

The user's JWT must include the universityId to allow the University Service to filter content automatically.

Verified students can see papers, notes, and solutions specific to their University.

University Service

1. Global Admin: Institutional Setup

User Story: As a Global Admin, I want to create a University and assign its primary Representative so that the institutional hierarchy can begin.

Acceptance Criteria:

The system must allow the creation of a unique University entity.

The admin must be able to assign a user as a UNIVERSITY_REP for a specific UNIVERSITY scope.

The admin must define global Exam Types (e.g., Midterm, EndSem) linked to the University ID.

2. University Representative: Department Management

User Story: As a University Representative, I want to add all departments and assign their respective leaders so that each academic wing is managed independently.

Acceptance Criteria:

The rep must be able to create Department records linked to their specific University.

The rep can assign a DEPARTMENT_REP to a specific DEPARTMENT scope.

Access must be restricted so a rep can only manage departments within their assigned University.

3. Department Representative: Program Coordination

User Story: As a Department Representative, I want to define the academic programs and assign Program Reps so that specific degrees (like B.Tech) are supervised.

Acceptance Criteria:

The rep must be able to create Program records (e.g., B.Tech CSE) linked to their Department.

The rep can assign a PROGRAM_REP to a specific PROGRAM scope.

The system must prevent a Department Rep from modifying programs in a different department.

4. Program Representative: Session Organization

User Story: As a Program Representative, I want to create academic sessions (batches) and assign Session Reps to manage the specific year-wise requirements.

Acceptance Criteria:

The rep must create an AcademicSession defining the start and end years for a batch.

The system must enforce a unique constraint on programId and startYear to prevent duplicate batches.

The rep can assign a SESSION_REP to a specific SESSION scope to manage a particular year/batch.

5. Session Representative: Content & Verification

User Story: As a Session Representative, I want to manage subjects and verify uploaded materials so that the paper repository remains accurate and high-quality.

Acceptance Criteria:

The rep must be able to add Subjects linked to a specific semesterNumber and AcademicSession.

The rep has the authority to change the status of an uploaded Paper to VERIFIED.

The rep must be able to manage and verify notes and solutions uploaded by other students within their specific session scope.

Paper/Content Service

1. Student: Paper & Note Upload

User Story: As a Verified Student, I want to upload a previous year's paper or my personal notes so that I can contribute to the university's resource pool.

Acceptance Criteria:

The user must be authenticated and associated with a specific University ID. The upload must require metadata: SubjectId, BatchId, ExamType, and Category (Paper/Note/Solution).

Files must be stored securely (e.g., AWS S3 or similar), and only the file metadata and URL are stored in the database.

Upon upload, the paper status is set to PENDING until a Session Rep verifies it.

2. User: Content Discovery (Search & Filter)

User Story: As a Student, I want to filter papers by Program, Session, and Subject so that I can find the specific study material I need for my exams.

Acceptance Criteria:

The system must allow filtering based on the University Service hierarchy: University -> Program -> Academic Session -> Semester -> Subject.

Users should only see content that has been marked as VERIFIED by a representative.

The search results should display the uploader's name (optional) and the upload date.

3. Session Representative: Content Validation

User Story: As a Session Representative, I want to review pending uploads for my assigned session so that I can ensure only high-quality, accurate papers are visible to students.

Acceptance Criteria:

The rep must see a "Review Queue" containing all PENDING uploads within their SESSION scope.

The rep must have the authority to APPROVE (set status to VERIFIED) or REJECT (delete or mark as BANNED) an upload.

If a paper is rejected, the rep should be able to provide a reason (e.g., "Wrong Subject," "Poor Quality").

4. System: Storage & Metadata Management

User Story: As a System, I want to link file metadata to the University hierarchy so that the content remains organized even as the university grows.

Acceptance Criteria:

Each paper record must store the uploadedBy User UUID for tracking.

The database must maintain a record of uploadedAt and updatedAt for every file.

User Service

1. Student: Profile Management

User Story: As a Student, I want to manage my personal profile information (name, profile picture, bio) so that I can maintain my identity on the platform.

Acceptance Criteria:

The user must be able to update their display name and profile image.

Profile data must be linked to the userId provided by the Auth Service.

The system should allow users to view their registration date and linked University.

2. Representative: Role & Authority Discovery

User Story: As a Representative (Univ/Dept/Prog/Session), I want to view my current active assignments so that I know exactly which academic scope I am authorized to manage.

Acceptance Criteria:

The system must return a list of all active RepresentativeAssignment records for the logged-in user.

Each assignment must clearly show the roleType (e.g., DEPT_REP) and the name of the scopeId (e.g., "Computer Science Dept").

The user should see the expiresAt date for temporary assignments.

3. Student: University Data Sync

User Story: As a Student, I want my profile to automatically reflect the University I selected during signup so that I don't have to manually search for it every time.

Acceptance Criteria:

Upon registration, the User Service must store the universityId as the user's primary "Home University".

The UI should automatically filter the dashboard based on this primary University ID.

4. System: Cross-Service Identity Resolution

User Story: As a System, I want to provide user metadata (name, email) to other services so that when a paper is uploaded, the "Uploaded By" field shows a real name instead of just a UUID.

Acceptance Criteria:

The User Service must provide an internal API for the Paper Service to fetch user details by userId.

The service must respect privacy settings, only sharing necessary metadata for verification logs.

Notification Service

1. System: OTP Delivery (Auth Integration)

User Story: As a System, I want to send an OTP to a user's email during registration so that they can verify their identity and activate their account.

Acceptance Criteria:

The service must receive a request containing an email address and a 6-digit OTP.

It must use the Resend API to deliver the email using a verified domain (e.g., noreply@pdeep.store).

The email must use a clean HTML template as defined in the technical requirements.

The service should log whether the email was "Sent Successfully" or if it failed.

2. User: Real-time Alerts for Representatives

User Story: As a Representative, I want to receive a notification when a student uploads a new paper in my session so that I can verify it promptly.

Acceptance Criteria:

When the Paper Service records a new upload, the Notification Service must trigger an alert to the assigned SESSION_REP.

The notification should contain a direct link to the "Pending Verification" dashboard.

The system should support multiple channels (Email initially, with hooks for In-App notifications later).

3. Student: Verification Updates

User Story: As a Student, I want to be notified when my uploaded paper has been verified or rejected so that I know the status of my contribution.

Acceptance Criteria:

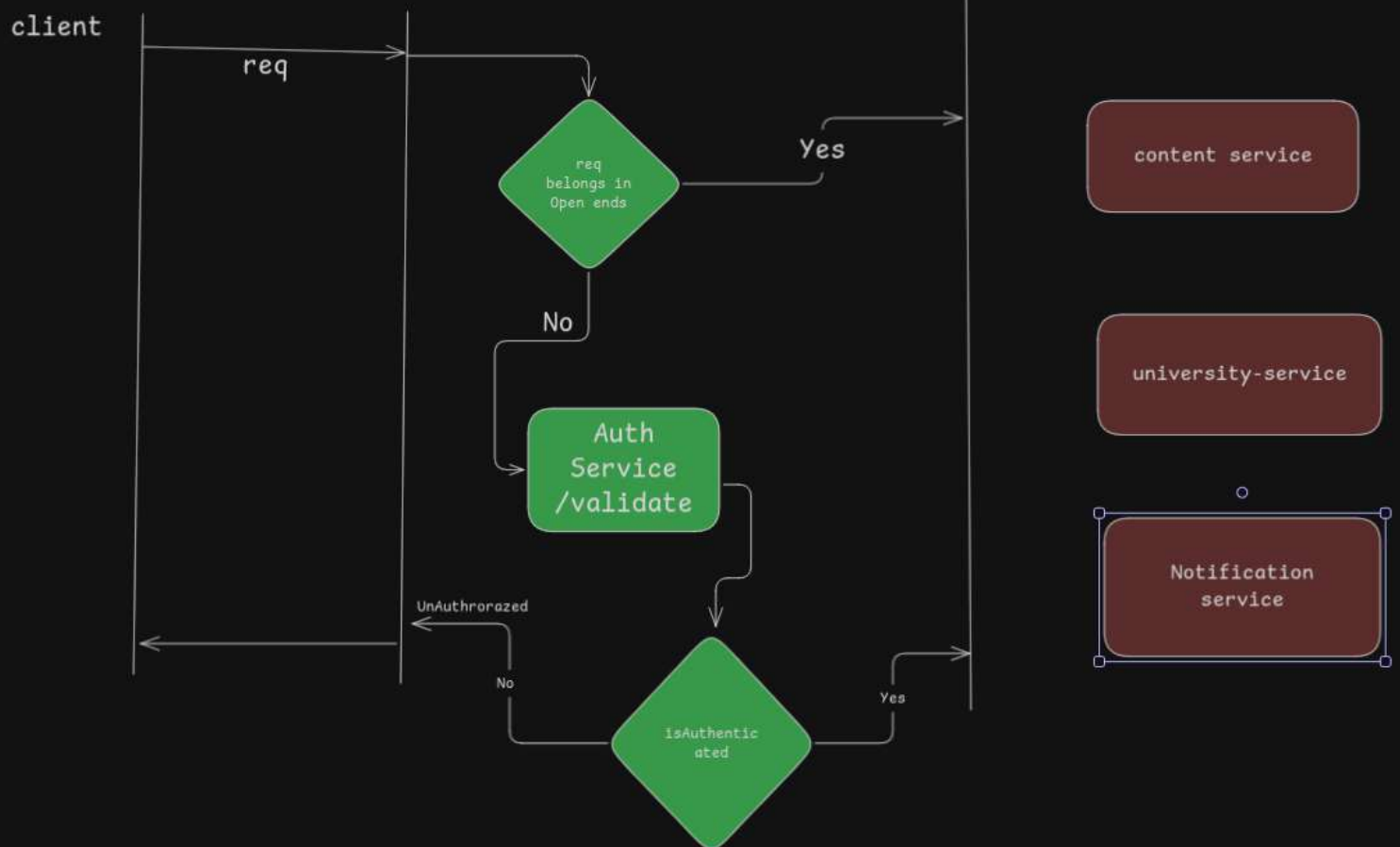
Trigger a notification when a Representative updates a paper status to VERIFIED or REJECTED.

If rejected, the notification must include the reason provided by the representative.

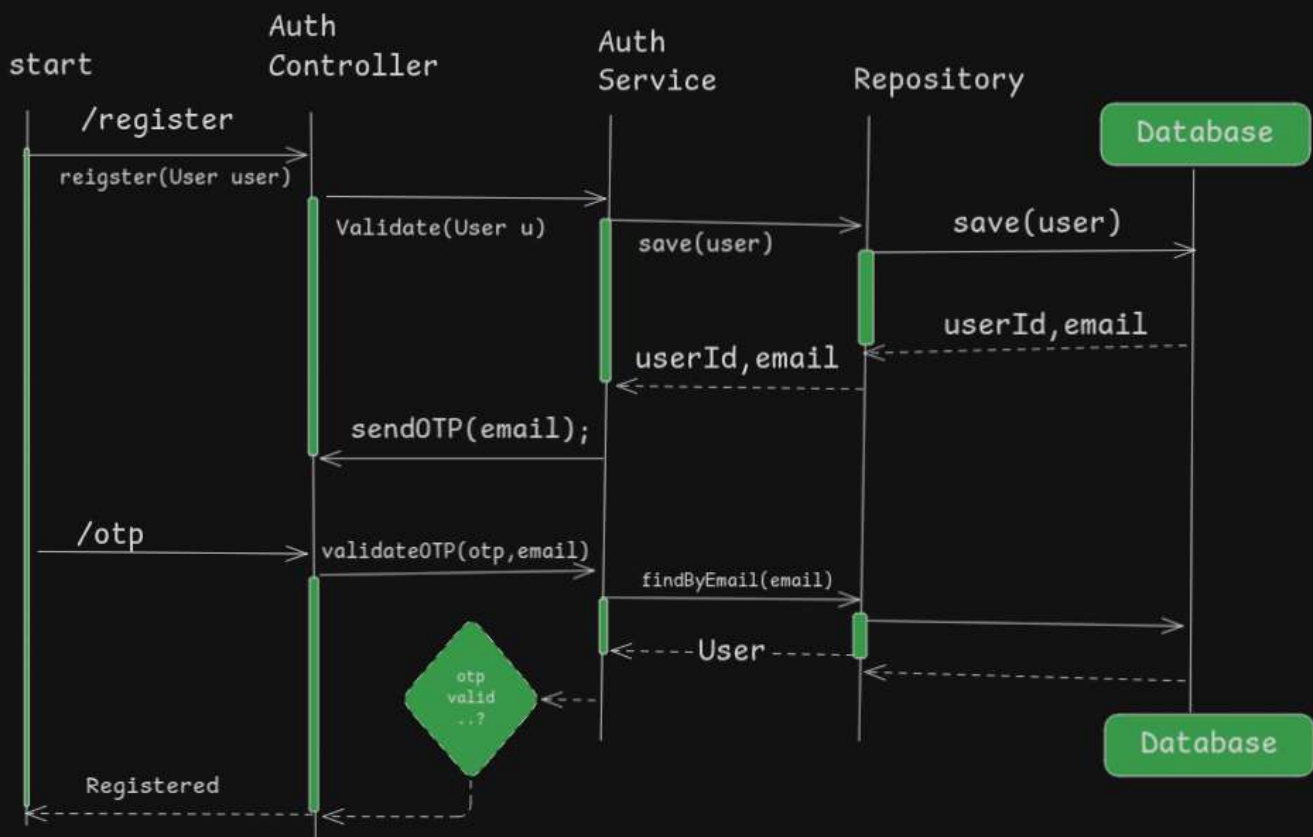
API-GATEWAY

API-GATEWAY

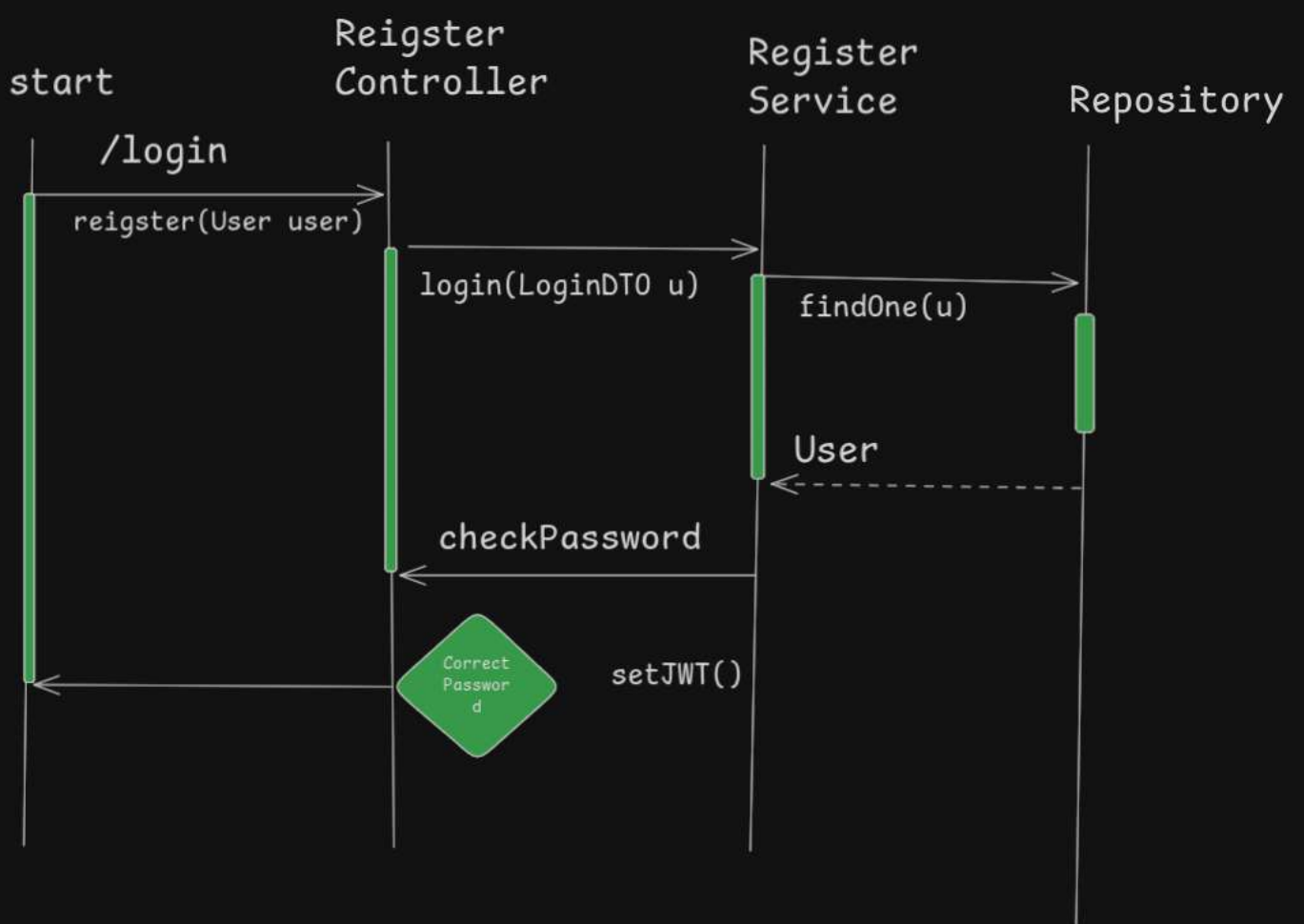
All Services



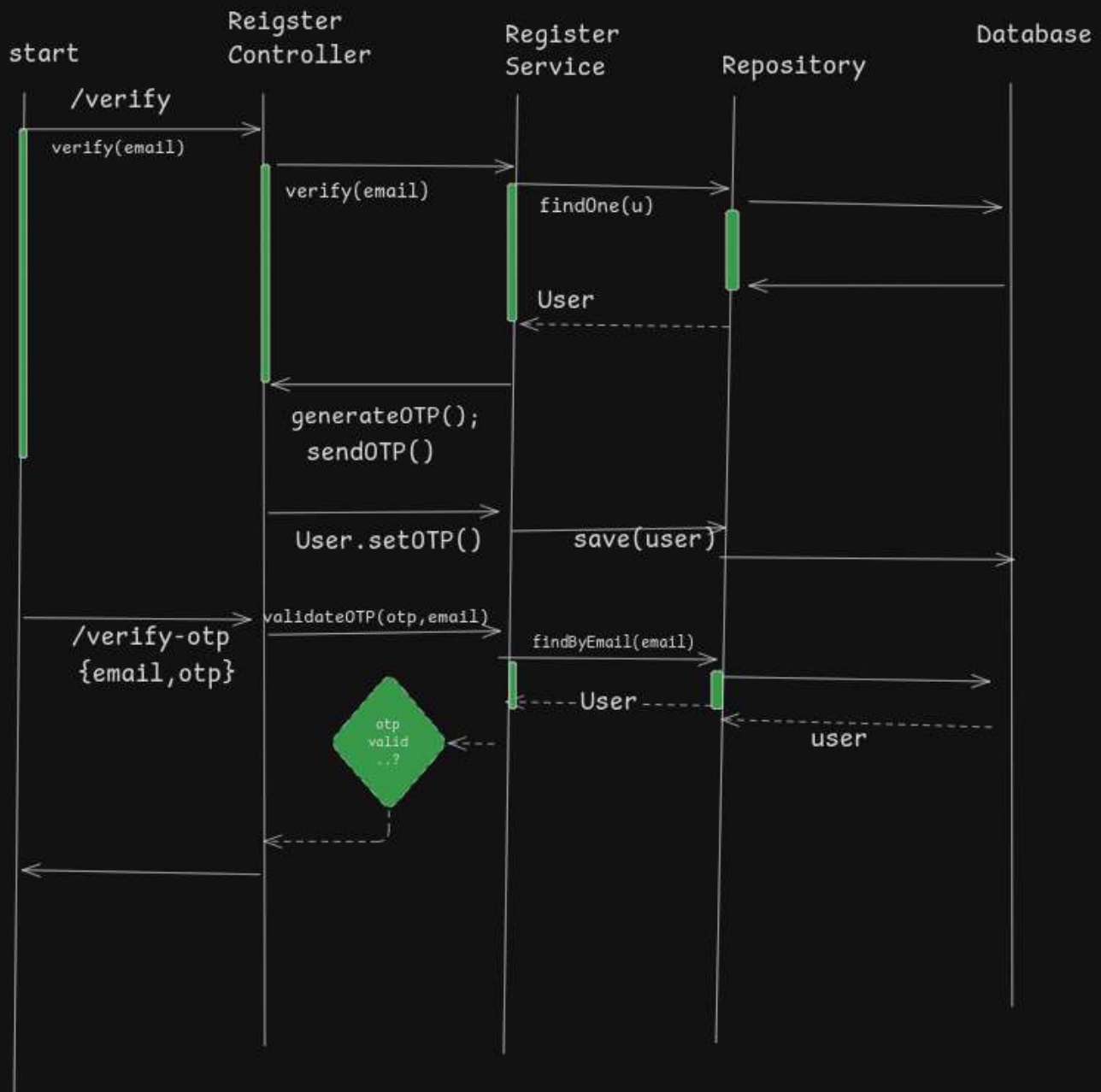
User Registration



User Login

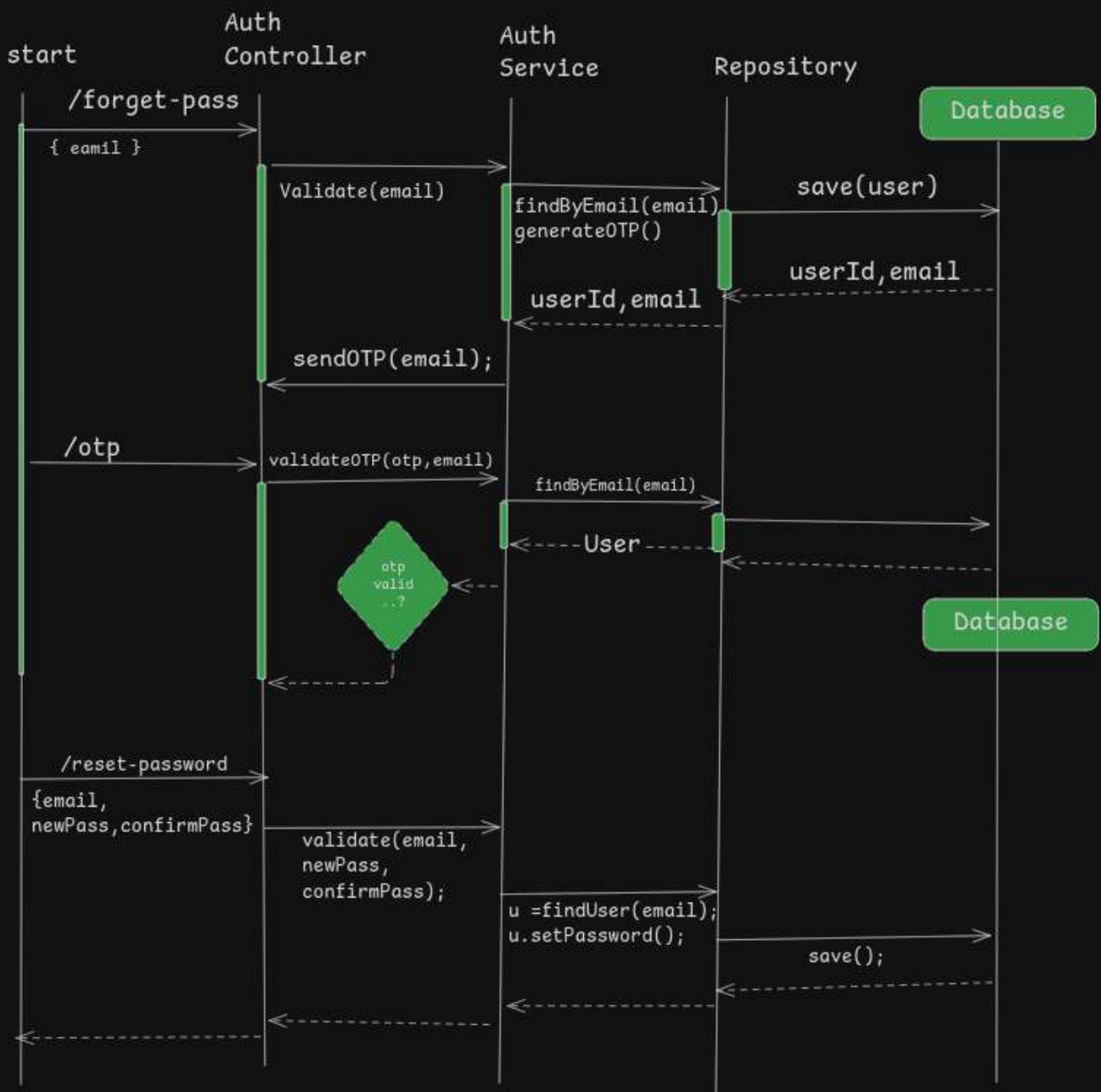


Email verification



To move canvas, hold [Scroll wheel] or [Space] while dragging, or use the hand tool

forget-passwrod
Reset password



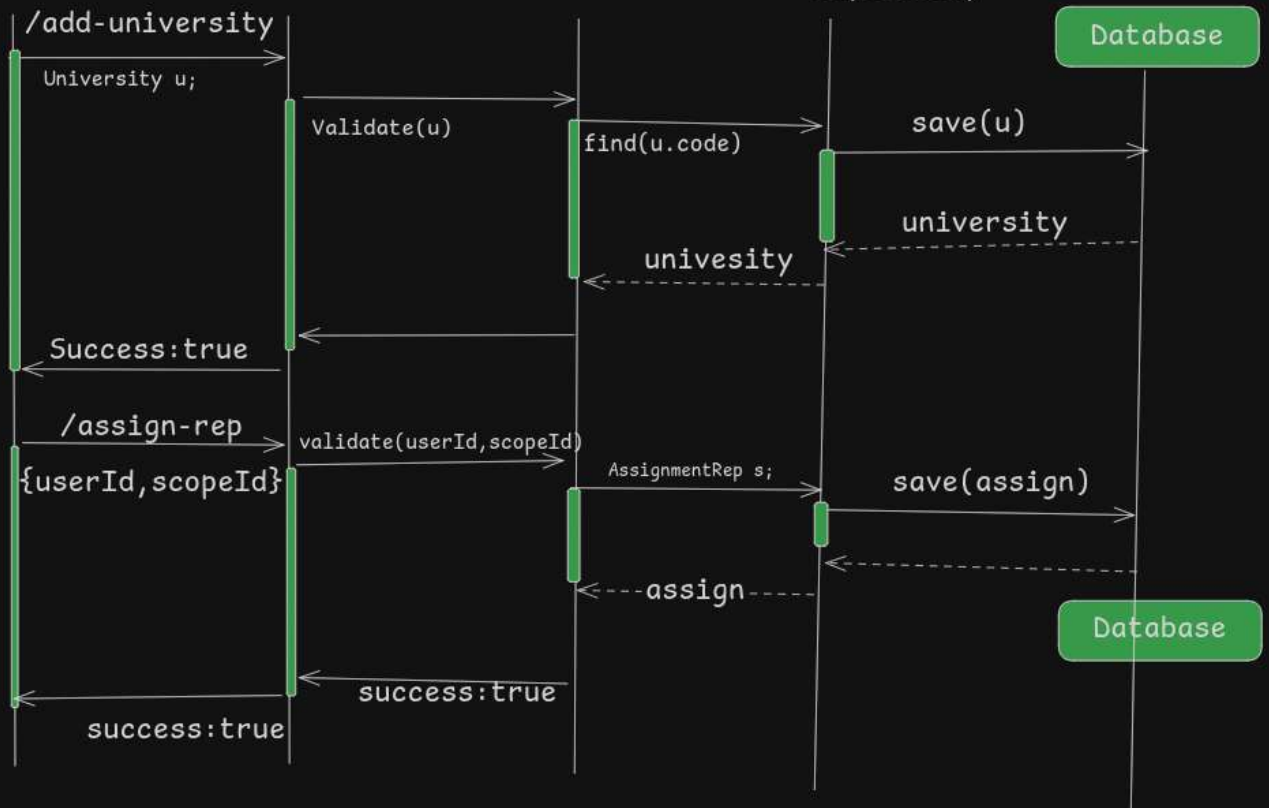
GlobalAdmin Dataflow

Add University

start

Repository

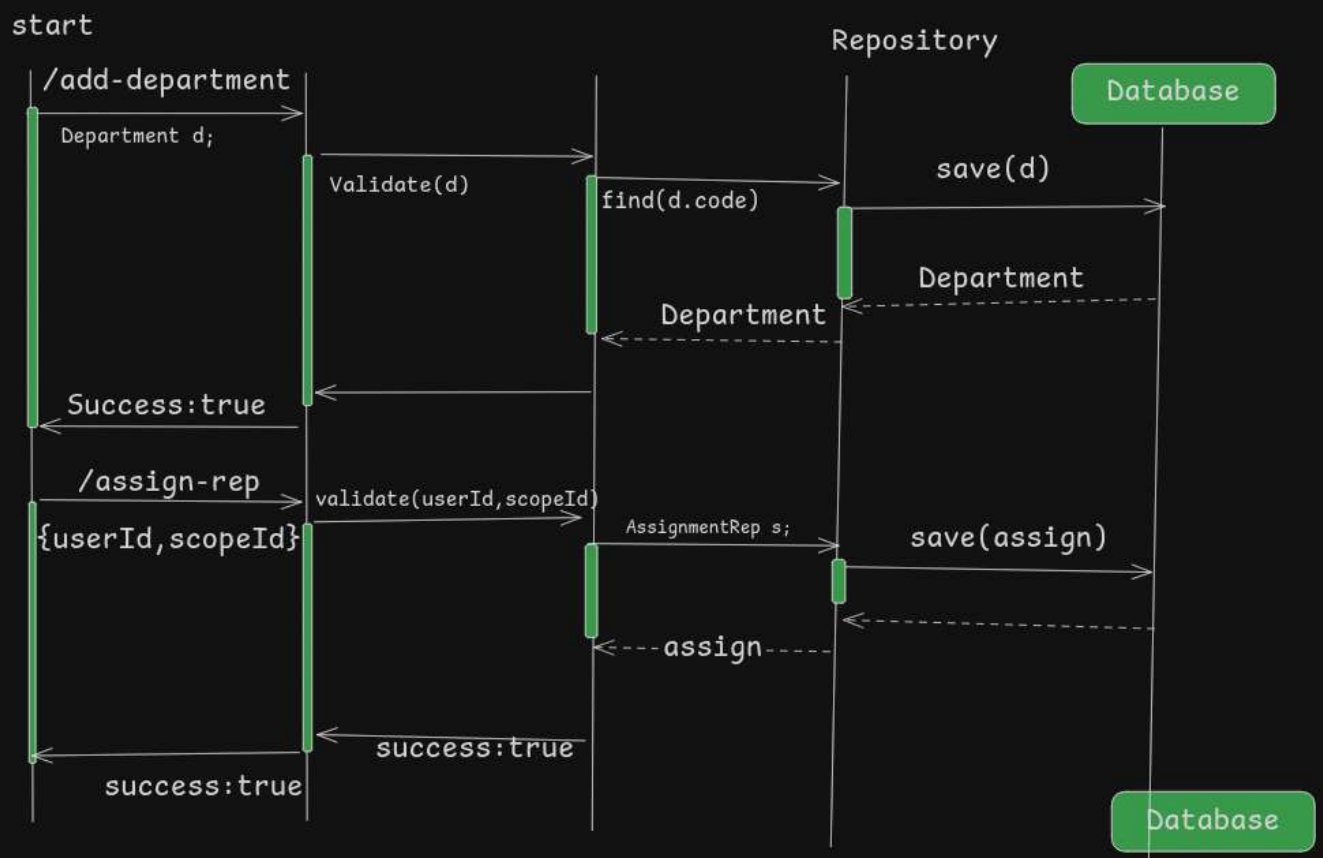
Database



Database

Department Dataflow

Add Department



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program Dataflow

Add Program

start

